

Main Branch Implementation Status

Prepared for: Internship Progress Review

Status date: 25 August 2026

Reference specifications: fcw_ttc_incremental_approach.pdf and fcw_ttc_alert_decision_from_height_rate.pdf

BASELINE	CURRENT MILESTONE	OVERALL ASSESSMENT
main commit d263bc2	Experiments 1-3 complete Experiment 4 partially complete	Integrated functional prototype Validation and stabilization remain

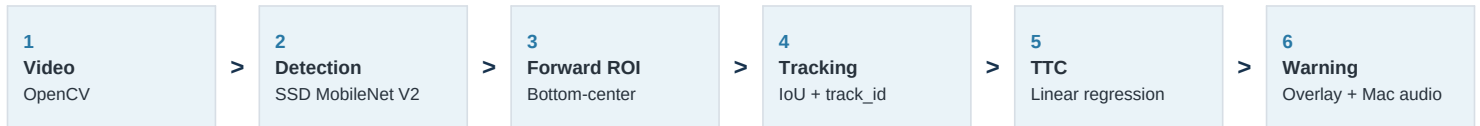
1. Executive Summary

The main branch contains an integrated recorded-video FCW pipeline: vehicle detection, forward-region filtering, per-vehicle tracking, TTC estimation from bounding-box height growth, hysteresis-based alert decisions, annotated video output, and macOS warning audio.

Current achievement: Experiments 1-3 meet their baseline requirements. Experiment 4 is partially complete through linear regression and alert hysteresis. The system should be classified as a functional integrated prototype, not yet as a quantitatively validated FCW implementation.

Main remaining gaps: three-consecutive-sample confirmation, TTC confidence validation, structured CSV logging, time-series plots, quantitative evaluation, headway correction, and more robust tracking.

2. Processing Flow



The estimator keeps a rolling height history and fits a straight line to height versus time. The fitted slope is dh/dt . TTC is calculated as $h/(dh/dt)$ only when the slope is positive. If any tracked vehicle is in the alert state, macOS plays Funk.aiff; audio is safely disabled on other operating systems.

3. Current Configuration

Parameter	Current value	Purpose
Detection class	COCO car = 3	Process cars only
Detection score	≥ 0.4	Minimum SSD confidence
IoU threshold	0.3	Track association threshold
Max missed frames	150	Track retention after missed detections
Height history	Maximum 10 / minimum 5	Rolling TTC estimation window
Alert ON	$TTC < 4.0$ s	Current experimental ON threshold
Alert OFF	$TTC > 6.5$ s	Hysteresis release threshold

4. Experiment Progress

Experiment	Specification objective	Main implementation	Status
1 Basic TTC	Height history -> dh/dt -> TTC -> alert	Rolling deque, linear-regression slope, approach-only TTC, TTC overlay.	Implemented
2 Forward ROI	Reject detections outside the ego-front region	Fixed trapezoid and OpenCV bottom-center point-in-polygon test.	Implemented
3 Tracking	Maintain object identity and history per vehicle	Greedy IoU association, ID creation, missed-frame retention, expired-state cleanup.	Baseline complete
4 TTC refinement	Filtering, regression, stable alerts, confidence checks, correction	Linear regression and ON/OFF hysteresis implemented. Confirmation counter, TTC confidence, and headway correction remain.	Partial

5. Component Status

File	Responsibility	Status
detector.py	TensorFlow SSD inference and car-score filtering	Complete
roi.py	Trapezoidal ROI and bottom-center filtering	Complete
tracker.py	IoU-based object IDs and track expiration	Baseline
ttc.py	Per-track deque, linear regression, dh/dt and TTC	Regression complete
alert.py	Per-track alert state and ON/OFF hysteresis	Partial
alarm.py	macOS afplay control; safe no-op on other OSs	Integrated extra
visualizer.py	ROI, ID, IoU, height, history, TTC and alert overlay	Complete
ssd_ttc.py	Pipeline orchestration, MP4 output and inference timing	Integrated

6. TTC Alert Decision Compliance

Specification item	Current main behavior	Assessment
N-frame height history	deque(maxlen=10); calculation starts at 5 samples	Implemented
Continuous recalculation	Regression and TTC are updated for every new detection	Implemented
Approaching target only	dh/dt <= 0 produces no TTC and disables alert	Implemented
Three consecutive samples	A single below-threshold TTC can enable the alert	Missing
Hysteresis	Alert ON below 4.0 s; OFF above 6.5 s	Implemented; values differ
Compare N=5 and N=10	Only one maximum-10/minimum-5 configuration is available	Not evaluated
Intermediate-value log	Console output only when TTC is available; no CSV	Partial
Height and TTC plots	Video overlay only; no time-series plots	Missing

7. Known Limitations

Threshold difference. The reference document proposes 2.0 s ON and 2.5 s OFF. Main currently uses experimental values of 4.0 s and 6.5 s.

Tracking. Greedy IoU matching can switch IDs when vehicles overlap, cross, or move rapidly. An ID switch contaminates the height history and therefore the TTC estimate.

TTC confidence. SSD confidence is filtered at 0.4, but main does not calculate regression R-squared, reject outliers, or apply an independent TTC confidence check.

Geometry. The fixed ROI is not exact ego-lane detection. The TTC is an image-scale estimate, not calibrated distance to the ego vehicle's front bumper.

8. Recommended Next Milestone

Priority	Work item	Expected result
1	Require three consecutive below-threshold TTC values	Fewer one-frame false warnings
2	Add CSV logging and height/TTC time-series plots	Repeatable quantitative comparison
3	Evaluate N=5 versus N=10 on the same videos	Evidence-based history-window choice
4	Add tracker tests and measure ID switches	Reliable per-vehicle height history
5	Add regression confidence and outlier rejection	More reliable TTC alerts

9. Operation

Command	Required local files
python code/ssd_ttc.py	model/ssd_mobilenet_v2_fpnlite_320x320_coco17_tpu-8/saved_model mp4/sample06.mp4

Conclusion: Main can execute the complete baseline FCW pipeline described by the reference documents. The next phase should prioritize warning-quality validation rather than adding more pipeline stages.